



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

A Mini Project Report

on

CAREVA: AI CAREER GUIDANCE PLATFORM

Submitted in partial fulfillment of the requirements for the course

“Introduction to Design Thinking”

(4th Semester, B. Tech – CSIT ‘A’)

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Abstract

This project presents Careva, a full-stack AI career guidance platform designed to help students identify suitable roles and prepare a focused roadmap. The system is developed using a React frontend, Flask backend, SQLAlchemy database models, and modular service logic for assessment, recommendation, resume scanning, progress tracking, and chat guidance.

The application collects learner inputs such as current skills, target domains, saved searches, career-test answers, resume evidence, completed courses, practiced skills, milestones, and project reviews. These inputs are processed through weighted role-matching logic to recommend suitable careers and explain missing skills.

The platform also includes advanced functionalities such as a learning lab, project-review workflow, progress tracker, profile management, resume vault, and contextual career coach. By integrating multiple career-readiness services in one interface, Careva provides a practical and personalized preparation experience for students.

Overall, Careva demonstrates how modern technologies like web development, recommendation algorithms, document analysis, and AI-assisted guidance can be effectively utilized to support student career planning and job readiness.

1 Problem Statement and Motivation

1.1 Problem Statement:

Many students struggle to choose a suitable career path because career information is scattered across job portals, course platforms, social media, online tutorials, college discussions, and generic counselling material. This creates confusion because the same student may receive different advice from different sources without knowing which direction truly matches their skills, interests, and long-term goals.

A learner may know a few skills, but may not understand which roles fit those skills, what is missing, which projects prove readiness, or how to convert learning into a job-ready roadmap. Students also find it difficult to compare role requirements, identify skill gaps, choose the right courses, build relevant portfolio projects, and track whether their preparation is improving over time.

Therefore, there is a need for a comprehensive and personalized system that can analyze student inputs, recommend suitable career roles, explain skill gaps, and guide practical preparation. Such a system should combine assessment data, resume evidence, saved interests, progress records, learning resources, and project feedback so that students can receive clear, actionable, and continuous career guidance from one platform.

1.2 Motivation:

The motivation behind Careva is to help students make informed career decisions through a simple and interactive platform that connects skills, interests, resumes, and learning progress. Instead of depending on random advice, students can use Careva to understand where they currently stand and what steps they should take next.

Career planning becomes easier when a student can view strengths, missing skills, roadmap resources, and portfolio requirements together instead of depending only on broad advice. This helps learners compare different career options more clearly and focus their time on skills, courses, and projects that actually support their chosen direction.

The project is motivated by the need for a student-friendly system that combines career testing, resume scanning, role recommendations, learning plans, project review, and chat guidance. By bringing these features into one platform, Careva reduces confusion and gives students a more organized way to prepare for future opportunities.

Ultimately, Careva aims to convert career confusion into a structured roadmap and demonstrate how modern web technologies can support real student readiness. The project shows how a web application can guide learners from self-assessment to practical preparation in a clear and useful manner.

2 Problem Approach / System Design

2.1.Overall Approach:

The development of Careva followed the five stages of Design Thinking:

1. **Empathize:** Studied students' challenges such as career confusion, unclear role requirements, scattered learning resources, weak resume feedback, and lack of personalized guidance.
2. **Define:** Identified the core problem as the absence of one platform that can analyze student skills, interests, aptitude signals, resume evidence, and progress data.
3. **Ideate:** Brainstormed features such as career-test scoring, role matching, resume skill extraction, learning lab resources, project review, progress tracking, and chat coaching.
4. **Prototype:** Developed a full-stack web application using React for the interface, Flask for the backend APIs, SQLAlchemy for data storage, and modular Python services for analysis.
5. **Test:** Validated the system through local execution, frontend production build, backend API testing, and live interface screenshots from the working Careva prototype.

2.2.Data Structures and Algorithms Used.

The system relies on structured database models, weighted scoring logic, keyword extraction, and API-based processing techniques:

- **Relational Database Schema (SQLAlchemy):** Utilizes structured models such as users, profiles, progress records, chat messages, career-test attempts, resumes, and project submissions.
- **Weighted Recommendation Algorithm:** Calculates a career match score using skill alignment, domain alignment, saved-search intent, assessment fit, and market demand.
- **Career Test Scoring:** Converts multiple-choice answers into aptitude scores such as analytics, communication, coding, design, and leadership with suggested domains.
- **Resume Skill Extraction:** Parses uploaded PDF or DOCX resumes, normalizes text, detects known skills using aliases, and improves recommendation evidence.
- **Rule-Based Chat Coach:** Uses profile data, progress details, role recommendations, and recent chat history to provide practical career guidance and next steps
- **Progress and Project Algorithms:** Tracks completed courses, practiced skills, milestones, reviewed projects, and portfolio readiness using structured scoring rules.

2.3. System Design

Careva utilizes a modular three-tier architecture to ensure scalability, maintainability, and efficient data processing. This architecture separates the user interface, backend logic, and database storage so that each part of the system can be developed, tested, and improved independently.

1. Client Layer (Frontend):

Built using React, Vite, and CSS, the frontend provides dashboard, career test, recommendation, learning lab, progress, profile, resume, and chat interfaces. It allows students to interact with the system easily, enter their details, view recommendations, track progress, upload resumes, and access career guidance through a clean and responsive interface.

2. Application Layer (Backend):

Developed using Flask, this layer handles authentication, session management, protected REST APIs, file uploads, recommendation logic, and chat processing. It acts as the main processing layer of the application by receiving requests from the frontend, validating inputs, applying business logic, and returning useful results to the user.

3. Database Layer:

Implemented using SQLAlchemy with SQLite fallback and SQL Server support, this layer stores accounts, profiles, resumes, progress, chats, assessments, and project details. It ensures that user data is stored in an organized manner and can be retrieved whenever required for recommendations, progress tracking, resume analysis, and personalized guidance.

3 Implementation

3.1.Problem Analysis and Description

To effectively implement Careva, the system had to address multiple student career-planning challenges, including unclear role selection, scattered learning resources, weak resume evidence, and limited progress visibility. Many students know they want to build a career in technology, but they often lack a clear method to connect their current skills with suitable roles and preparation steps.

This was achieved by designing a multi-module web application that integrates assessment scoring, recommendation logic, resume parsing, project review, chat coaching, and progress tracking. Each module handles a specific part of the career-guidance process, while the complete system works as one connected platform for student career readiness.

The application collects inputs such as current skills, target domains, saved searches, career-test answers, resume evidence, completed courses, practiced skills, and milestones. These inputs help the system understand the learner's profile more accurately and generate recommendations that are more relevant to their goals.

All components work together to provide personalized career matches, skill-gap explanations, roadmap resources, portfolio requirements, and next-step guidance for students. This helps learners move from general confusion to a clear preparation plan that they can follow and improve over time.

3.2.Modules Identified.

- 1. Authentication Module:**
Handles user registration, login, logout, session storage, and protected access to user-specific APIs.
- 2. Career Test Module:**
Displays multiple-choice questions and converts answers into aptitude scores and suggested career domains.
- 3. Recommendation Module:**
Ranks career roles based on current skills, target domains, saved searches, market demand, and assessment-fit logic.
- 4. Resume Vault Module:**
Allows users to upload PDF or DOCX resumes, extracts readable text, identifies skills, and updates recommendation evidence.
- 5. Learning Lab Module:**
Provides roadmap videos, course links, project briefs, project-review upload, latest feedback, and progress-ring visualization.
- 6. Progress Tracker Module:**
Records completed courses, practiced skills, milestones, weekly focus, streak days, and reviewed projects.
- 7. Chat Coach Module:**
Generates contextual career replies using learner profile data, recommendations, progress status, and recent chat history.
- 8. Profile Module:**
Allows students to maintain education level, preferred work style, current skills, saved searches, and target domains.

3.3.Full Source Code (Core Algorithmic Logic with Comments)

1. # 1. ROLE RECOMMENDATION SCORING ALGORITHM

```
# Calculates career match percentage for each role

def score_role(profile, role):

    current_skills = normalize(profile['current_skills'])

    target_domains = normalize_ranked(profile['target_domains'])

    scores = profile['assessment']['scores']

    # Skill alignment rewards existing role-related skills

    skill_alignment, matched, missing = weighted_skill_alignment(
        current_skills, role['skills']
    )

    # Domain and search alignment connect learner intent

    domain_alignment = domain_alignment_score(target_domains, role['domains'])

    search_alignment, search_hits, intent = search_alignment_score(
        profile['saved_searches'], role
    )

    # Fit alignment compares test scores with role weights

    fit_alignment = cosine_similarity(
        vector_from_scores(scores), vector_from_role(role)
    )

    market_alignment = role['demand_score'] / 100

    match_score = min(100, round(
        skill_alignment * 31 + domain_alignment * 15
        + search_alignment * 14 + fit_alignment * 24
        + market_alignment * 10 + intent * 6
    ))

    return match_score, matched, missing
```

2. # 2. CAREER TEST SCORING ALGORITHM

```
# Converts selected answers into aptitude scores

def score_career_test(answers):
```

```

scores = defaultdict(int)
domain_weights = Counter()

for question in CAREER_TEST_QUESTIONS:
    selected = find_selected_option(question, answers)
    score_weight = QUESTION_SCORE_WEIGHTS[question['id']]
    domain_weight = QUESTION_DOMAIN_WEIGHTS[question['id']]

```

3. RESUME SKILL EXTRACTION ALGORITHM

```

# Extracts readable resume evidence for matching

for axis, value in selected['scores'].items():
    scores[axis] += value * score_weight

for domain in selected['domains']:
    domain_weights[domain] += domain_weight

normalized_scores = {
    axis: min(10, max(1, round(4 + scores[axis] * 1.2)))
    for axis in ['analytics', 'communication', 'coding', 'design', 'leadership']
}

return normalized_scores, top_domains(domain_weights)

```

```
def analyze_document(path, filename):
```

```

    text = extract_text(path)

    if not text:
        return limited_scan_result(filename)

    normalized = normalize_text(text)

    matched_skills = []

```

4. PROGRESS SNAPSHOT AND PROJECT GATE

```

# Ensures full completion requires portfolio proof

for skill in all_catalog_skills:
    aliases = {skill, *SKILL_KEYWORD_ALIASES.get(skill, [])}

```

```

    if any(has_keyword(text, normalized, alias) for alias in aliases):
        matched_skills.append(skill)

    matched_domains = collect_resume_domains(text, normalized, matched_skills)
    profile = build_profile_from_resume(text, matched_skills, matched_domains)
    recommendations = recommendation_engine.build_recommendations(profile)[:5]
    return scan_summary(matched_skills, matched_domains, recommendations)

```

3. # 3. PROGRESS SNAPSHOT AND PROJECT GATE

Ensures full completion requires portfolio proof

```

def build_progress_snapshot(progress):
    completed = len(progress['completed_courses'])
    practiced = len(progress['practiced_skills'])
    milestones = len(progress['milestones'])
    project_reviews = len(progress['project_reviews'])
    raw_score = min(100, completed * 10 + practiced * 6 + milestones * 6)
    raw_score += min(project_reviews, 3) * 18
    raw_score += round(average_project_score(progress) * 0.22)
    remaining_projects = max(0, 3 - project_reviews)
    project_gate = 100 if remaining_projects == 0 else 35 + round((project_reviews / 3) * 65)
    return min(raw_score, project_gate)

```

4. # 4. REST API ROUTE STRUCTURE

```

GET /api/health          -> service and storage status
POST /api/auth/register  -> create account and session
POST /api/auth/login     -> authenticate learner
GET /api/profile         -> saved profile, test, resumes
POST /api/career-test/submit -> score test and refresh recommendations
GET /api/recommendations -> ranked roles and upskill plan
POST /api/resume/upload   -> scan resume and update evidence
POST /api/project/review  -> review portfolio artifact

```

4 Output and Results

4.1 Sample Input and Output

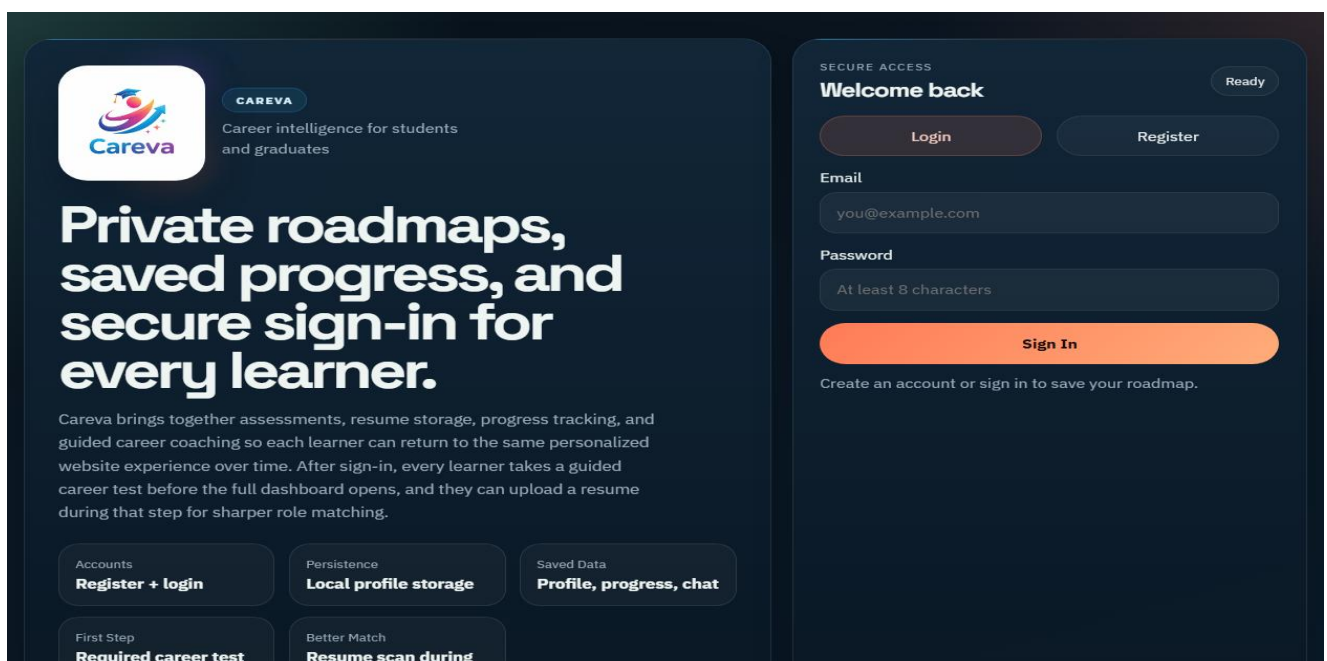
The system was validated through representative student career-guidance workflows:

1. **Input (User Registration):** The learner enters name, email address, and password.
Output: Careva creates an authenticated session and initializes profile and progress data.
2. **Input (Career Test):** The learner answers questions about role family, coding comfort, tools used, subject strength, weekly upskilling time, and blockers.
Output: Careva generates aptitude scores and suggested domains.
3. **Input (Resume Upload):** The learner uploads a text-based PDF or DOCX resume.
Output: Careva extracts matched skills, detects possible domains, and updates recommendation evidence.
4. **Final Output (System Response):**
 - Careva displays ranked career roles with match percentage, matched skills, missing skills, domain fit, and recommended next steps.
 - Careva shows learning resources, project briefs, project-review feedback, progress score, completed courses, practiced skills, and milestones.
 - Careva provides contextual chat guidance for resumes, interview preparation, skill gaps, projects, and role-specific learning plans.
 - The system also maintains profile details and saved searches so recommendations remain personalized across sessions.

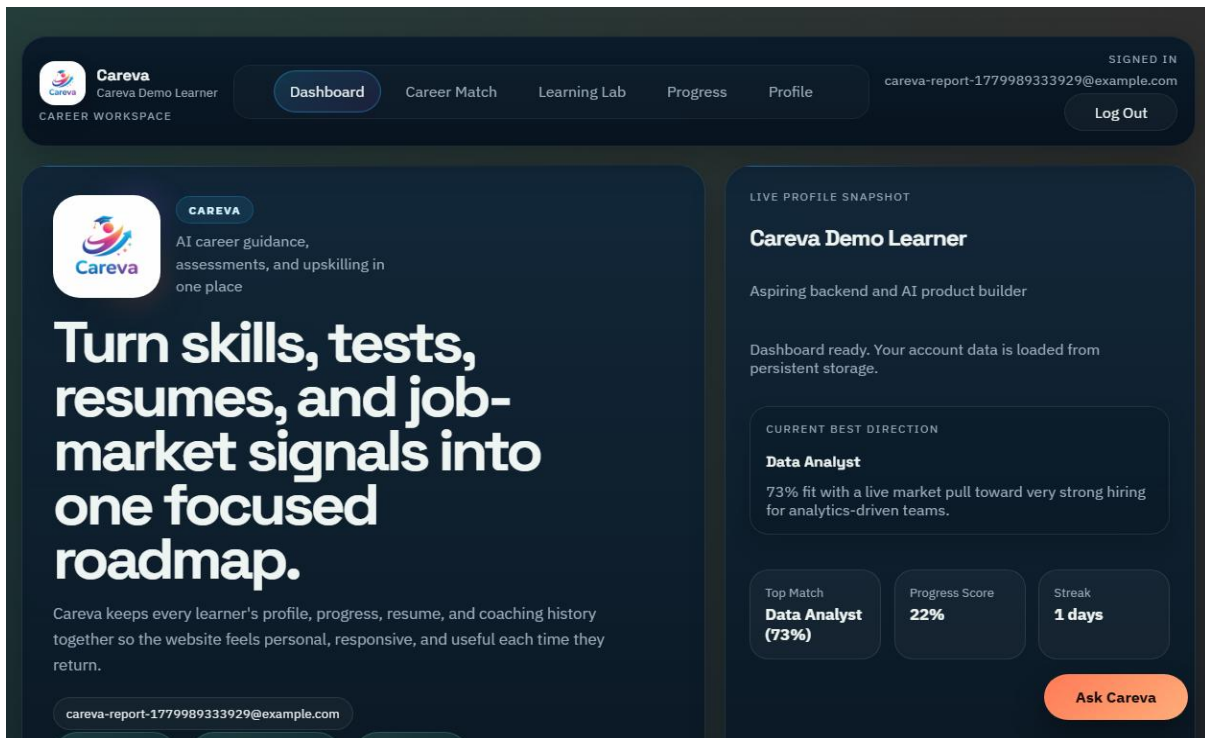
4.2 Screenshots / Output Evidence

The UI prototype of Careva ensures a focused, student-friendly career-planning experience while separating complex backend processing from the frontend interface.

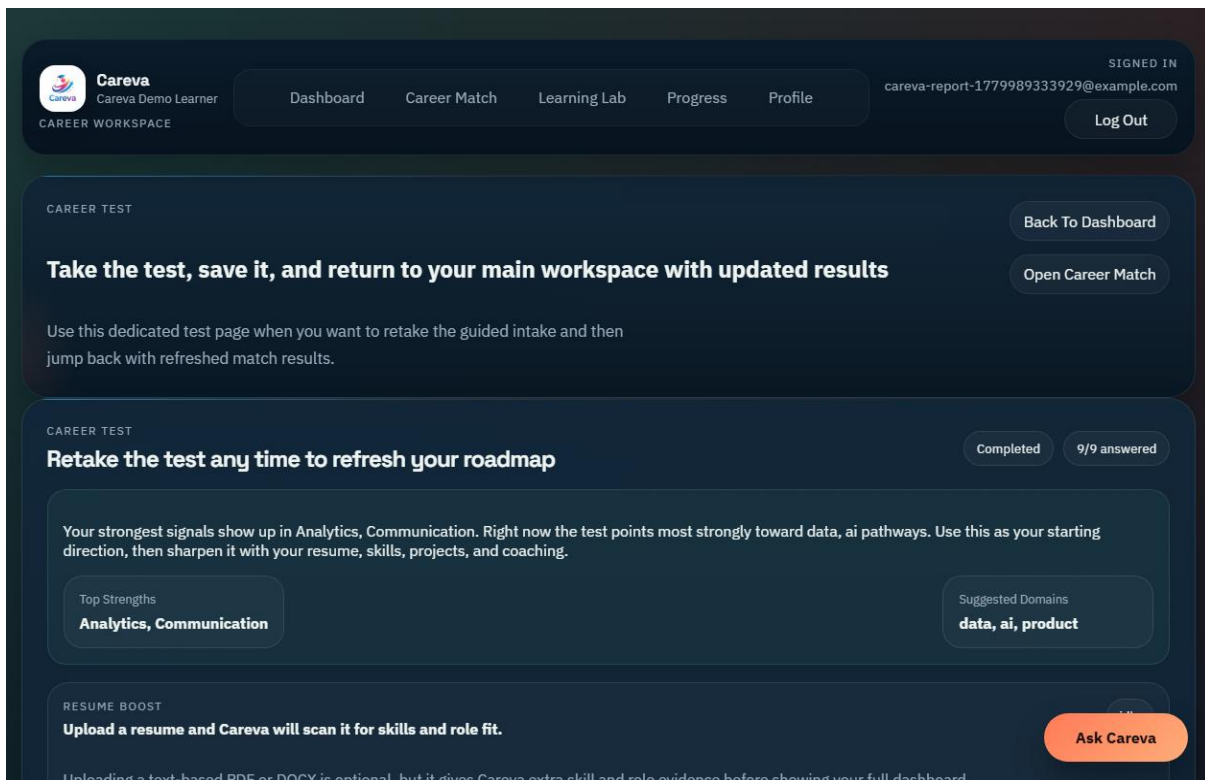
- **Login Screen:** A clean authentication page where learners create an account or sign in securely before accessing the personalized workspace.



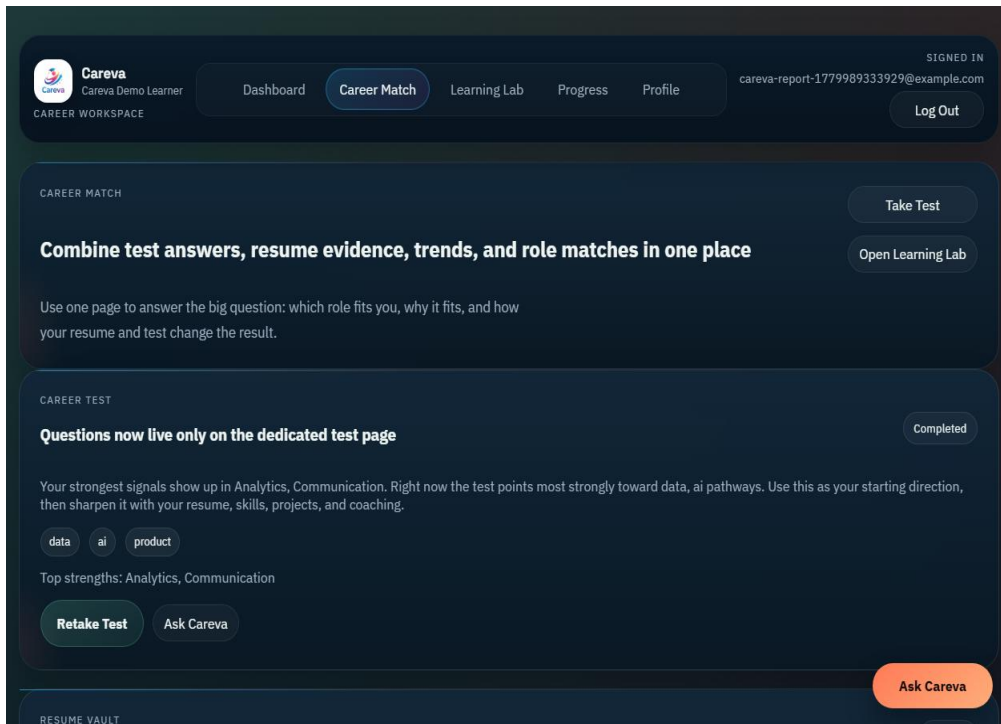
- **Home Dashboard:** The central interface showing best career direction, top role match, profile snapshot, progress score, streak, and quick actions.



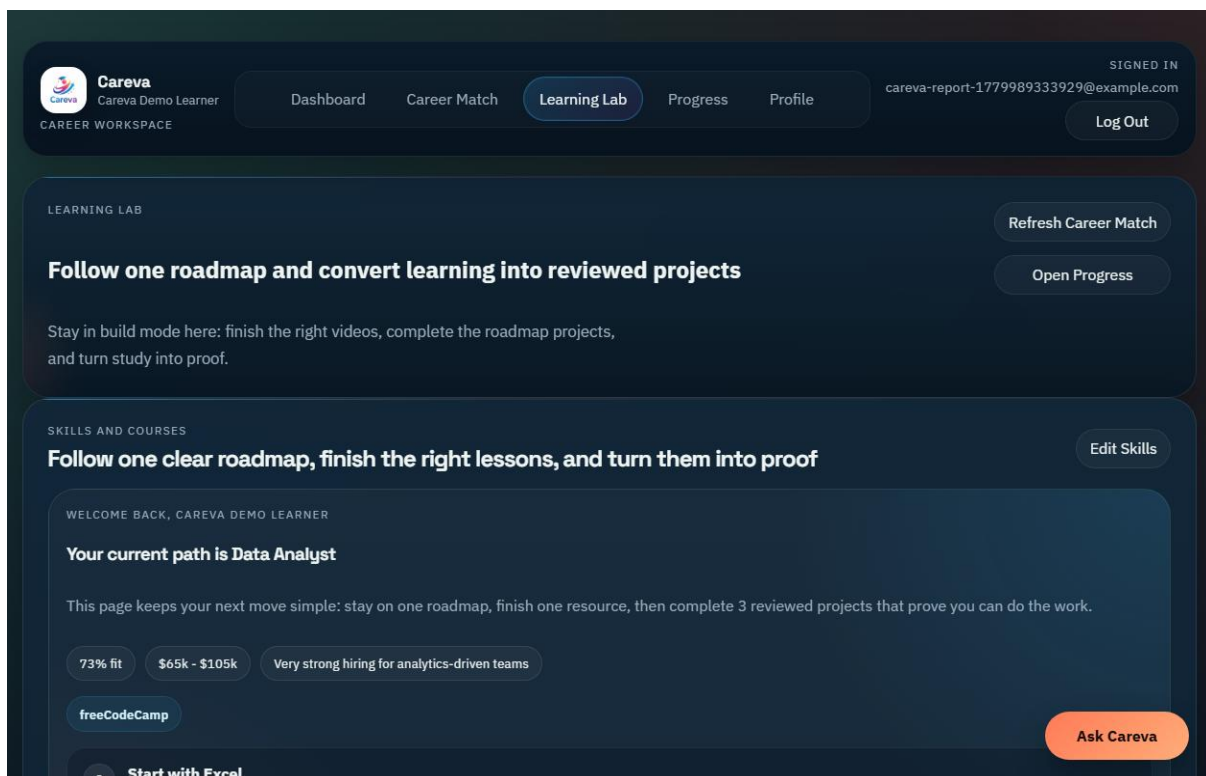
- **Career Test Page:** Allows users to answer role-fit questions and optionally upload resume evidence to improve career recommendations.



- **Career Match Page:** Displays market pulse, recommendation cards, match reasoning, skill gaps, and role-specific preparation suggestions.



- **Learning Lab Page:** Provides roadmap videos, course links, required project briefs, project-review upload, feedback, and progress-ring status.



- **Progress Tracker Page:** Shows completed courses, practiced skills, milestones, weekly focus, streak days, and the three-project portfolio requirement.

Careva
Careva Demo Learner
CAREER WORKSPACE

SIGNED IN
careva-report-1779989333929@example.com
Log Out

Dashboard Career Match Learning Lab **Progress** Profile

PROGRESS Open Learning Lab Review Career Match

Track progress from courses, projects, milestones, and weekly focus

Keep all momentum signals together here so the learner can see what is finished, what is blocked, and what still needs proof.

PROGRESS TRACKER 22% complete

Keep momentum visible

Courses **1** Skills Practiced **1** Milestones **1** Project Reviews **0**

CURRENT FOCUS

Choose a direction and start a focused learning plan

Weekly focus: Build one API project and improve the resume proof section

PORTFOLIO GATE

Project requirement: 0/3 reviewed • 3 left

Project reviews will start shaping this quality signal once you upload

LIVE ACTIVITY Real-time feel

What changed recently

- Focused role: Data Analyst**
73% match with key gaps in Excel, Power BI, Statistics.
- Career test completed**
Your strongest signals show up in Analytics, Communication. Right now the test points most strongly toward data, ai pathways. Use starting direction, then sharpen it with skills, projects, and coaching.

Ask Careva

- **Profile Page:** Allows users to edit education level, preferred work style, current skills, saved searches, and target domains.

Careva
Careva Demo Learner
CAREER WORKSPACE

SIGNED IN
careva-report-1779989333929@example.com
Log Out

Dashboard Career Match Learning Lab Progress **Profile**

PROFILE Open Career Match Open Learning Lab

Manage your account and saved learner profile

Keep account details and editable learner preferences together in one calmer personal space.

ACCOUNT PROFILE

Edit your saved learner profile

Careva Demo Learner

These profile fields are yours to edit. Careva should adapt to what you write here, not invent it for you.

Dashboard ready. Your account data is loaded from persistent storage.

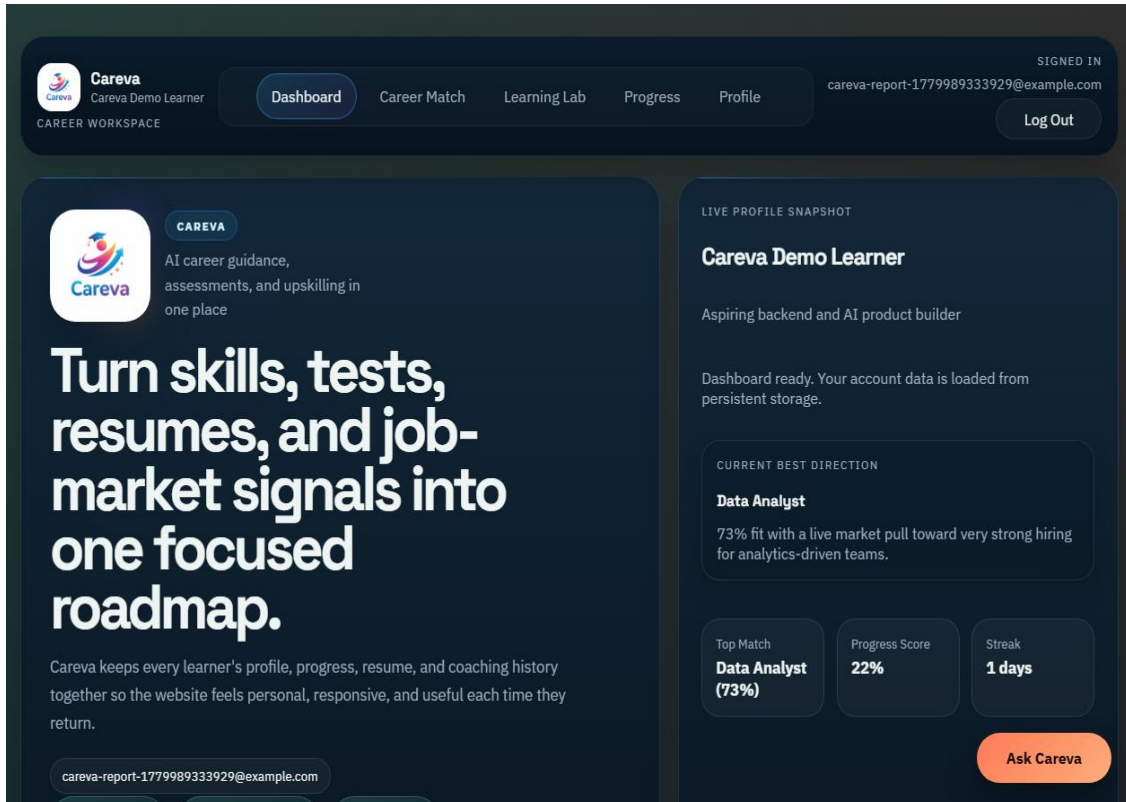
Progress **22%** Resume Files **0** Career Test **Completed**

EDITABLE FIELDS **ROADMAP PREFERENCES**

Name

Ask Careva

- **Career Coach Context:** The dashboard and coach access help learners receive guidance based on their profile, progress, and recommended roles.



4.3 Result Discussion and Analysis

The prototype evaluation confirmed that Careva successfully meets its design objective of providing personalized career guidance for students through one integrated platform. The system brings together career assessment, role recommendations, resume analysis, learning resources, progress tracking, and chat guidance in a single workspace.

The weighted recommendation model gives explainable results because each role match is supported by skill alignment, domain fit, saved-search intent, market demand, and assessment scores. This makes the recommendations easier for students to understand, since they can see why a role is suggested and what skills they need to improve.

The resume scan improves personalization by identifying evidence directly from uploaded documents, while the project-review module encourages portfolio-based readiness. These features help the system move beyond general advice and provide guidance based on the learner's actual skills, documents, and project work.

Testing across workflows such as registration, assessment, recommendation, learning lab, progress tracking, profile editing, and screenshots demonstrated that the interface is usable and connected. Each main section of the application worked as part of the same career-guidance flow, allowing students to move from input to output without confusion.

Overall, the results indicate that Careva is efficient, student-friendly, and capable of converting scattered career information into a practical roadmap. The prototype shows that a structured digital platform can support students in understanding their career direction and preparing for it step by step.

5 Conclusion and Learning Outcomes

5.1 Conclusion

The Careva project successfully demonstrates the effective use of modern technologies such as React, Flask, SQLAlchemy, resume parsing, recommendation logic, and AI-assisted guidance to solve a student career-planning problem.

The integration of career-test scoring, role recommendation, resume scanning, learning resources, project review, progress tracking, and chat coaching ensures that students receive structured and practical guidance.

Overall, the project proves that personalized career guidance can be delivered through a simple, user-friendly, and intelligent platform that helps learners plan and improve their readiness.

5.2 Learning Outcomes

Developing this project provided valuable technical and practical insights:

- **Design Thinking:**
Gained experience in understanding real student career-planning problems and designing a user-friendly solution.
- **Recommendation Logic:**
Learned how to build weighted scoring based on skills, domains, saved searches, market demand, and assessment fit.
- **API Integration:**
Acquired skills in building and consuming REST APIs between a React frontend and Flask backend.
- **Backend Development:**
Developed proficiency in Flask routing, session handling, file uploads, service modules, and application logic.
- **Database Management:**
Improved knowledge of SQLAlchemy models for storing users, profiles, resumes, chats, progress, assessments, and projects.
- **System Integration:**
Learned to combine assessment, recommendation, resume analysis, progress tracking, and chat coaching into one cohesive system.

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